

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Polynomials

Polynomials become manageable when roots, factors, and remainders are translated correctly.

POLYNOMIALS · P2 · 02

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The Map

This strategy booklet was mined from the local topic problem/answer PDFs, Dr Eslam's source notes, and the WMA12 website answer bank. The topic reduces to these exam moves, with hard variants kept visible on each page.

01 REMA: Remainder Theorem

TRIGGER *remainder when divided by $ax - b$*

02 FACT: Factor Theorem

TRIGGER *show $ax - b$ is a factor*

03 DIV: Divide And Reduce

TRIGGER *factorise cubic, quotient, repeated roots*

04 ROOT: Build From Roots

TRIGGER *given roots, form polynomial*

BANK EVIDENCE Local pages: 18 problem, 34 answer, 15 notes. Website primary entries: 15.

01 REMA: Remainder Theorem

TRIGGER *remainder when divided by $ax - b$*

BECOMES Substitute the value that makes the divisor zero.

FIRST LINE TO WRITE

$$\text{remainder} = f\left(\frac{b}{a}\right)$$

SIMPLEST STRATEGY

- 1 Read the divisor.
- 2 Equate it to zero.
- 3 Make the substitution.
- 4 Answer the remainder.

WORKED MODEL

| For $2x - 3$, use $x = \frac{3}{2}$, so the remainder is $f(\frac{3}{2})$.

HARD VARIANTS

- 1 If the remainder is given, set $f(b/a)$ equal to that value and solve for the parameter.
- 2 Use exact fractions, not decimals, for the substitution.
- 3 If the divisor is quadratic, use division or factor it first.

— BOTTOM LINE

For $ax - b$, the input is b/a .

02 FACT: Factor Theorem

TRIGGER *show $ax - b$ is a factor*

BECOMES A zero remainder proves the factor.

FIRST LINE TO WRITE

$$f\left(\frac{b}{a}\right) = 0$$

SIMPLEST STRATEGY

- 1 Find the zero of the divisor.
- 2 Apply the substitution.
- 3 Check it gives zero.
- 4 Tell the factor conclusion.

WORKED MODEL

| If $f(2) = 0$, then $x - 2$ is a factor.

HARD VARIANTS

- 1 For parameter questions, use the zero condition before factorising fully.
- 2 After one factor is known, divide to reduce degree.
- 3 Do not confuse root $x = a$ with factor $x + a$.

— BOTTOM LINE

Zero remainder means exact factor.

03 DIV: Divide And Reduce

TRIGGER *factorise cubic, quotient, repeated roots*

BECOMES Write polynomial as factor times quotient plus remainder.

FIRST LINE TO WRITE

$$f(x) = (x - a)q(x) + r$$

SIMPLEST STRATEGY

- 1 Decide the divisor.
- 2 Insert quotient terms.
- 3 Value-match coefficients or divide.
- 4 Identify the quotient.
- 5 Drop the remainder if it is zero.
- 6 Extract remaining roots.

WORKED MODEL

$$x^3 - 6x^2 + 11x - 6 = (x - 1)(x^2 - 5x + 6).$$

HARD VARIANTS

- 1 Comparison of coefficients is safer when the divisor has unknowns.
- 2 If the leading coefficient is not 1, match it before finding roots.
- 3 Repeated roots appear as repeated factors after division.

— BOTTOM LINE

Cubic questions usually become quadratic after one factor is removed.

04 ROOT: Build From Roots

TRIGGER *given roots, form polynomial*

BECOMES Each root gives one factor.

FIRST LINE TO WRITE

$$x = a \Rightarrow (x - a)$$

SIMPLEST STRATEGY

- 1 Record the roots.
- 2 Obtain factors.
- 3 Organise product.
- 4 Tune the leading coefficient.

WORKED MODEL

| Roots 2, -3 give $k(x - 2)(x + 3)$.

HARD VARIANTS

- 1 If a point on the curve is given, use it to find the multiplier.
- 2 Complex or surd roots may arrive in conjugate pairs.
- 3 Check multiplicity when the graph touches the axis.

— BOTTOM LINE

Roots and factors are the same information in different clothes.

Quick Reference

TRIGGER → FIRST ACTION

TRIGGER	FIRST ACTION
remainder when divided by $ax - b$	$\text{remainder} = f\left(\frac{b}{a}\right)$
show $ax - b$ is a factor	$f\left(\frac{b}{a}\right) = 0$
factorise cubic, quotient, repeated roots	$f(x) = (x - a)q(x) + r$
given roots, form polynomial	$x = a \Rightarrow (x - a)$